Updating Wi-Fi drivers and disabling power saving

Article KB0010022 | Owner: Network Support | Subcategory: Wi-Fi | Applies to: Dell Latitude 5440/7440, Lenovo ThinkPad T14/X1 Carbon, HP EliteBook 840, Any Windows 11 23H2/24H2 laptop with Intel, Realtek or Qualcomm Wi-Fi adapter

Summary

This article guides users and service-desk agents through troubleshooting intermittent Wi-Fi connectivity on Acme Global laptops by updating the wireless network adapter driver and disabling power-saving features that can cause drops. It also covers verifying the machine certificate used for 802.1X authentication and confirming a stable connection to the ACME-Corp SSID.

Symptoms

- Wi-Fi keeps disconnecting or drops intermittently
- Cannot connect to office Wi-Fi (ACME-Corp) despite correct credentials
- Wi-Fi shows connected but no internet access
- Wi-Fi is very slow in meeting rooms or crowded areas
- Corporate Wi-Fi repeatedly asks for username/password

Common causes

- Outdated or corrupted wireless network adapter driver
- Adapter power-saving settings that turn off the radio to save energy
- Expired or corrupted machine certificate used for 802.1X authentication
- Interference or channel congestion in high-density areas
- Misconfigured Wi-Fi profile causing repeated re-authentication

Preliminary Checks

Before proceeding with driver updates or power-setting changes, verify basic connectivity and rule out simple issues. Confirm that the laptop is within range of an ACME-Corp access point and that other devices can connect successfully. Check that the Wi-Fi adapter is enabled and not in airplane mode.

1. 1. Click the network icon in the taskbar and ensure ACME-Corp appears in the list of available networks.
2. Attempt to connect to ACME-Corp; note any error message (e.g., "Cannot connect to this network" or "Authentication failed").
3. If connection succeeds but there is no internet, open a browser and try to reach https://it.acme.example to confirm portal access.
4. If the SSID does not appear, toggle Wi-Fi off/on or restart the laptop.

Self-Help Procedure: Update Wi-Fi Driver and Disable Power Saving

Follow these steps to download the latest driver from the vendor's support site (or via Windows Update) and then adjust the adapter's power-management options. This resolves most cases where the radio powers down intermittently or the driver fails to handle 802.1X re-authentication correctly.

1. 1. Open Settings -> Windows Update -> Check for updates. Install any pending updates and restart if prompted.
2. If Windows Update does not offer a newer driver, identify the adapter:
   a. Right-click Start -> Device Manager.
   b. Expand Network adapters and note the exact model (e.g., Intel Wi-Fi 6 AX201).
   c. Visit the vendor's support page (Dell, Lenovo, HP) and download the latest Windows 11 driver for that model.
   d. Run the installer and follow the prompts; restart when finished.
3. After the driver is installed, open Device Manager again, right-click the Wi-Fi adapter -> Properties.
4. Select the Power Management tab.
5. Uncheck the box Allow the computer to turn off this device to save power.
6. Click OK, then close Device Manager.
7. Re-connect to ACME-Corp and monitor the connection for at least 15 minutes.

Service Desk Procedure (Network Support)

When a user reports persistent Wi-Fi drops after attempting self-help, the Network Support team should verify driver version, certificate status, and power-settings remotely or via guided steps. The procedure below outlines the actions to take and the information to collect for escalation if needed.

1. 1. Ask the user to reproduce the issue and capture the exact error message or event timestamp.
2. Request the user to open Device Manager and provide:
   a. Adapter name and driver version.
   b. Date of driver installation.
3. Using Intune or a remote session, check the driver version against the latest approved version in the Acme Global driver repository.
   a. If outdated, push the approved driver package via Intune or instruct the user to reinstall from the vendor site.
4. Verify the machine certificate used for 802.1X:
   a. Open certmgr.msc -> Personal -> Certificates.
   b. Locate the certificate issued by Acme Global CA; note the expiry date.
   c. If expired or within 30 days of expiry, initiate a certificate renewal request through the IAM team.
5. Confirm power-setting is disabled:
   a. In Device Manager -> Adapter Properties -> Power Management, ensure "Allow the computer to turn off this device to save power" is unchecked.
6. After making changes, ask the user to reconnect to ACME-Corp and perform a speed test (e.g., fast.com) and a ping test to gateway (ping -n 20 gateway_ip).
7. Document the pre- and post-change driver version, certificate expiry, and power-setting state in the ticket.
8. If the problem persists after driver update, certificate renewal, and power-setting adjustment, escalate to the Wireless/RF specialist within Network Support with the captured logs (netsh wlan show all, eventvwr -> System -> Wireless-LAN).

Verification and Testing

After completing the driver update and power-setting changes, verify that the Wi-Fi connection remains stable under typical usage conditions. Perform both quick checks and a longer stress test to ensure the issue is truly resolved before closing the ticket.

1. 1. Confirm the laptop shows "Connected, secured" to ACME-Corp with full signal bars.
2. Open a browser and navigate to the internal IT portal (https://it.acme.example) - it should load without prompting for credentials.
3. Run a continuous ping to the default gateway for 5 minutes: ping -t <gateway_ip>. Look for "Request timed out" - there should be none.
4. Launch a video-conference call or stream a high-bandwidth video for at least 10 minutes; note any drops or re-authentication prompts.
5. If all tests pass, close the ticket with the resolution description: "Updated Wi-Fi driver to version X.X.X, disabled adapter power saving, renewed machine certificate (if applicable), verified stable connection to ACME-Corp."

When to escalate

Escalate to the Network Support - Wireless/RF specialist when: (a) the Wi-Fi adapter driver is already at the latest approved version and power-saving is disabled, yet disconnections continue; (b) the machine certificate cannot be renewed or shows repeated validation failures; (c) packet captures reveal frequent de-authentication frames or RADIUS timeout errors. Include in the escalation: adapter model and driver version, certificate expiry date, netsh wlan show all output, relevant event-viewer logs (System -> Wireless-LAN), and a summary of troubleshooting steps already performed.

Keywords: Wi-Fi disconnect, ACME-Corp keeps asking for password, wireless driver update, disable Wi-Fi power saving, 802.1X certificate renewal, Wi-Fi slow in meeting room, corporate wireless drops, Wi-Fi no internet, Wi-Fi troubleshooting